



DIGITAL TECHNOLOGIES

ZUKUNFT. DIGITAL. NACHHALTIG.

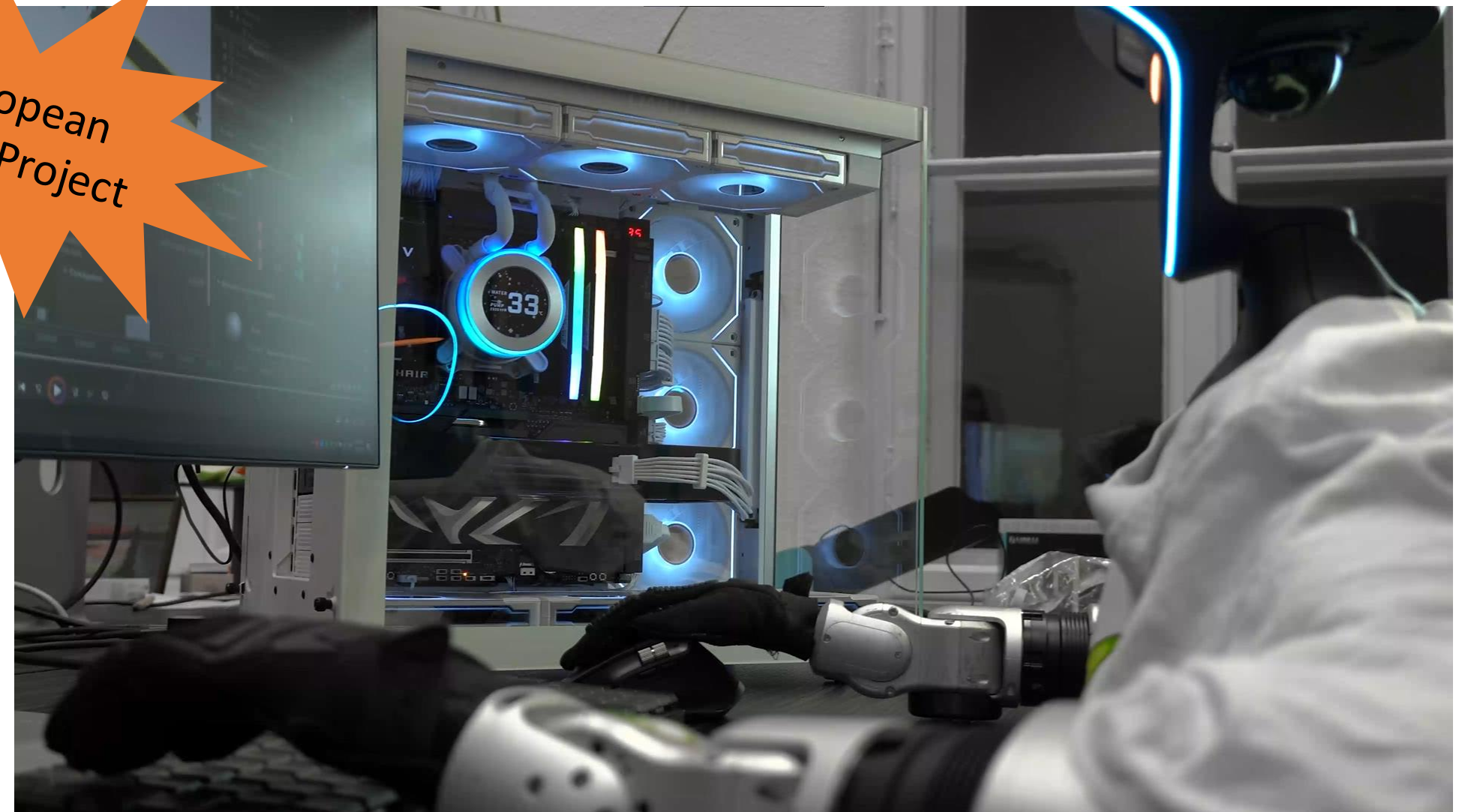
WiSe 2026

European
Team Project

TEMPO

Trained Engine for Memory-Paced
Openings

Machine Learning and COgnitive SoftwaRE (CORE)
Research Group
AI Team Projects (ATP)

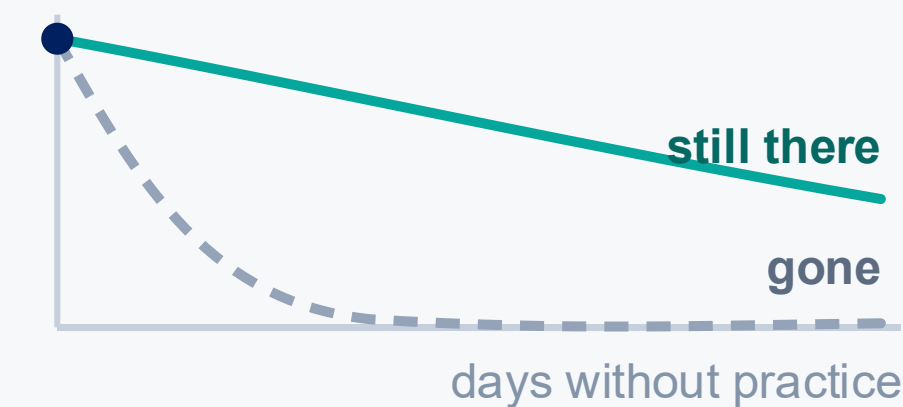


Ein gemeinsames Studienprogramm der

THE IDEA

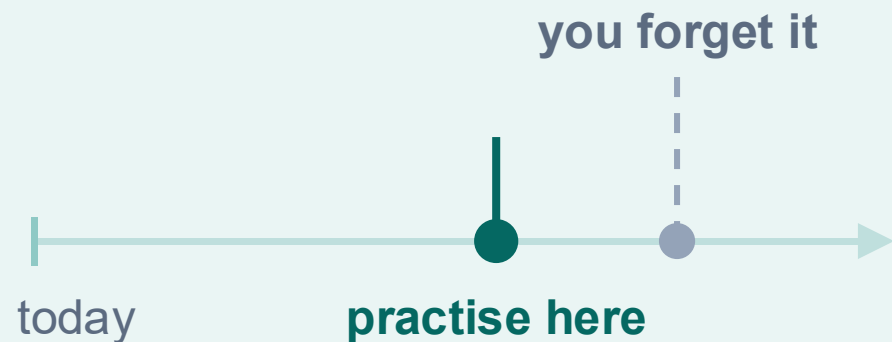
A model of a user's memory

Everyone forgets on their own schedule



Same learner, same day, two pieces of material.

Let's build something that predicts when you forget



The review planner turns that prediction into your next practice sessions.

Your practice is the training signal

what you got right

what you missed

how long you hesitated



Base model, fine-tuned on you

THE DOMAIN

Chess openings are where the model is built and measured first

Opening

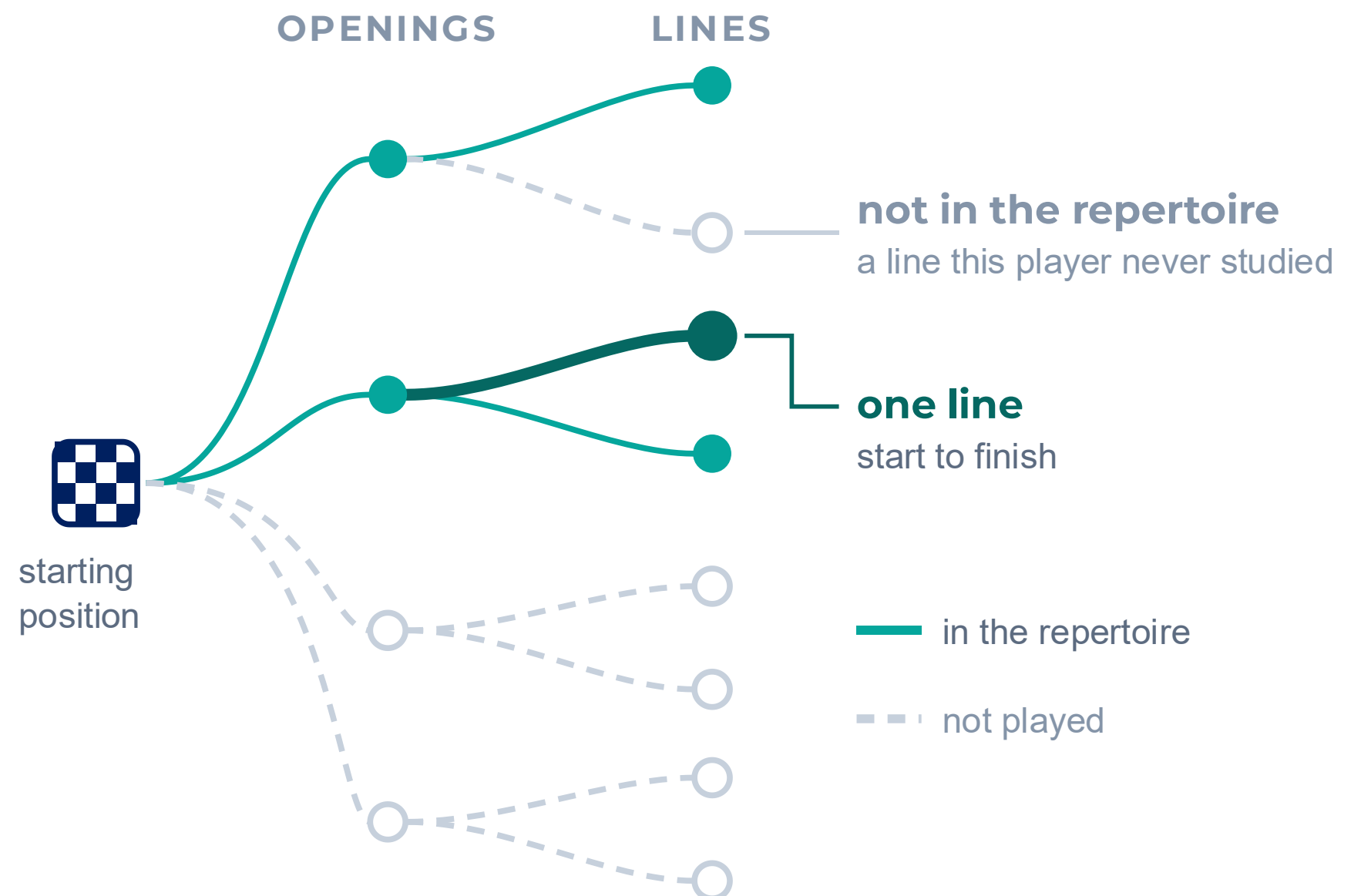
The first phase of a game, the part players memorise in advance.

Line

One concrete move sequence through an opening. This is the unit TEMPO reviews and drills.

Repertoire

The openings a player has chosen to play, and the specific specific lines within them they keep in memory.

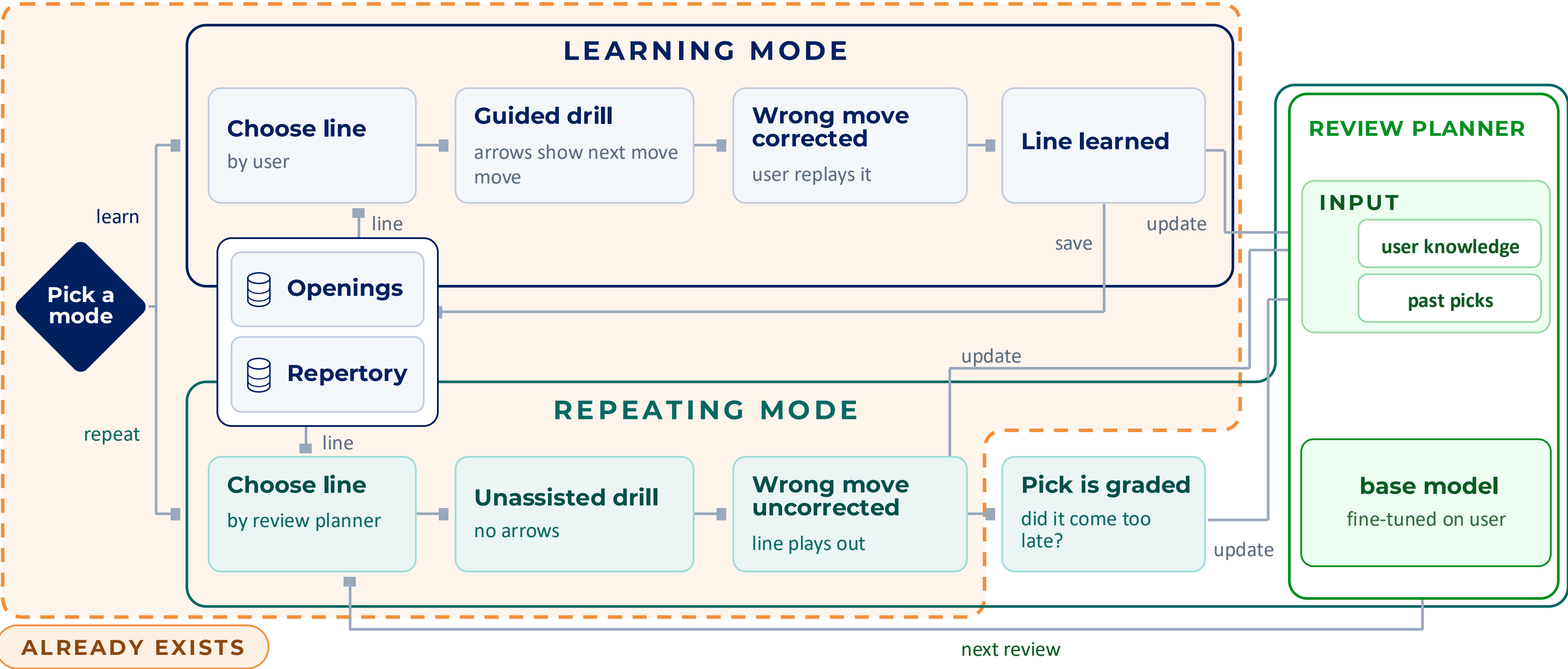


Existing chess trainers never learn whether they were right

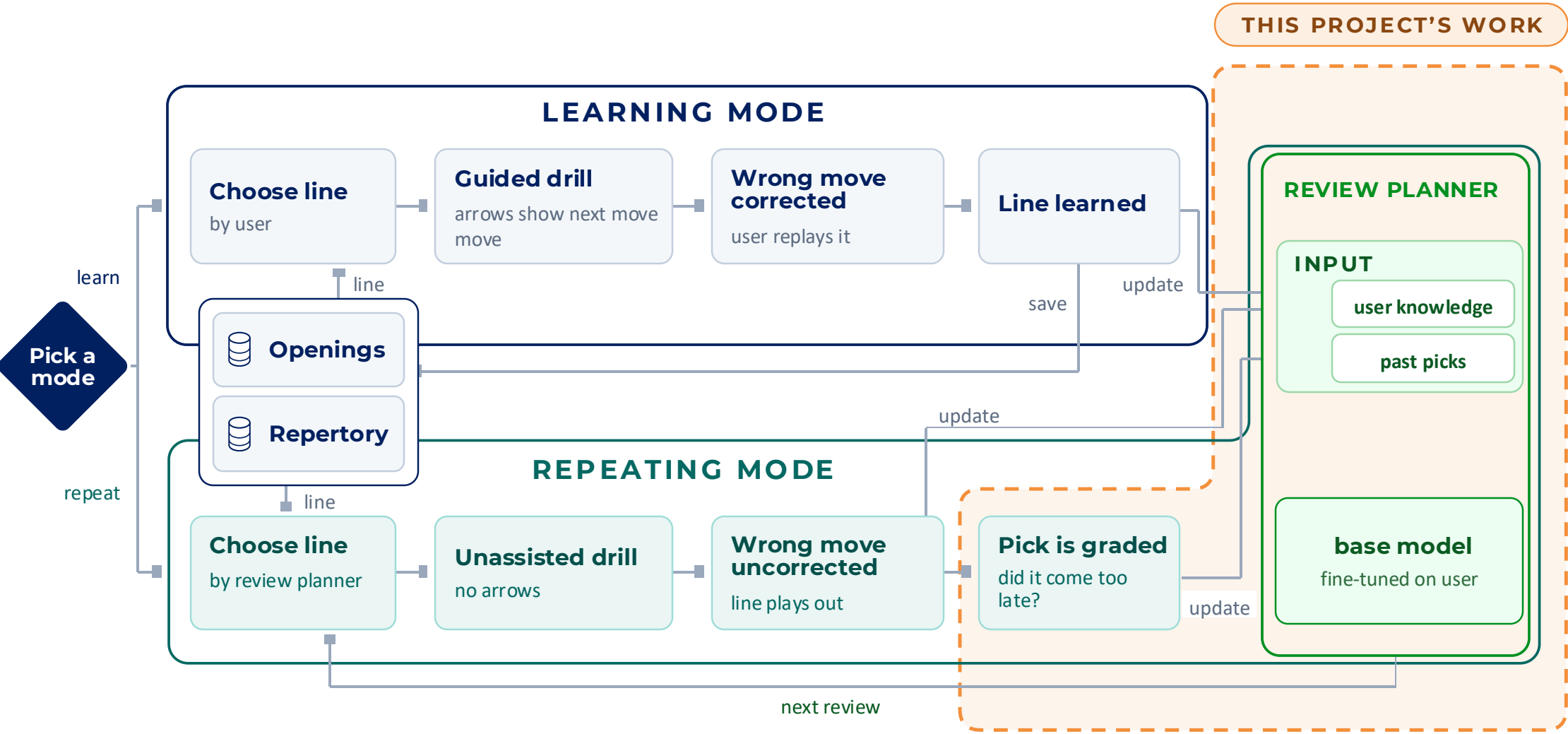
	OTHER CHESS TRAINERS	TEMPO
Who decides what to repeat	One fixed scheduling rule	A model fine-tuned on your way of memorizing memorizing
How much it adapts	Identical for every player, fixed for the life of the the product	One model with different weights for every user, user, updated after every session
Feedback on its own choices	None — a scheduler is never told whether it was it was right	The review planner’s picks are graded

HOW IT RUNS

Two modes, one record of what you know



How the review planner learns what to pick



- 1 Ships as a base model.**
One model out of the box, then fine-tuned on one player's own way of memorizing.
- 2 Every session is a label.**
Every move attempt and every review is logged.
- 3 Three inputs, one prediction.**
User repertoire, per-line practice history and the the model's own prior reviews go in; a single item comes out: the next line to repeat.
- 4 The target is recall timing.**
Each pick is scored against observed recall in that that session: what was recalled, what was missed, missed, and latency to the correct move. A line already forgotten counts as a late review.

Over time the model builds a picture of one specific player's memory.

Thinking about joining?

WHICH COMPETENCES ARE NEEDED?

- You are comfortable writing Python
- You are comfortable with git and working in a shared repository
- Interest in machine learning — recommender systems, personalization, or learning from user feedback
- Curiosity about how people learn and forget — this is a memory model before it is a chess model
- Chess knowledge is welcome for intuition, and is not required for most of the codebase

WHAT DO YOU LEARN IN THIS PROJECT?

- Fine-tuning a base model on one individual user's data
- Building a recommender that is graded on its own decisions and on user performance
- Designing the logging and evaluation that such a model is trained on
- Working in a mixed international team, across two universities and one shared repository

Be part of an international team!

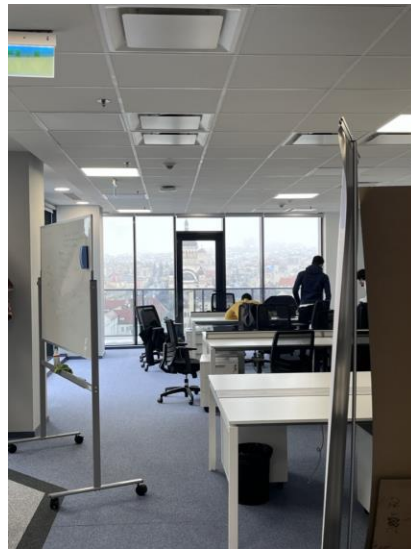
together with students from Babeş-Bolyai University, Cluj, Romania

For more info
and videos
visit:



<https://www.core-network.ai/ai-team-projects>

Intensive Week in Cluj



Social Events



SAVE THE DATE

Intensive Week Cluj:

~ 11.11. – 21.11.2026

Costs for flight and accommodation will be covered

Intensive Week Goslar:

~ 01.02. – 07.02.2027

CORE Projects in WiSe26

LECTRA

FastMile

Anbietende



UNIVERSITATEA
BABEȘ-BOLYAI

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